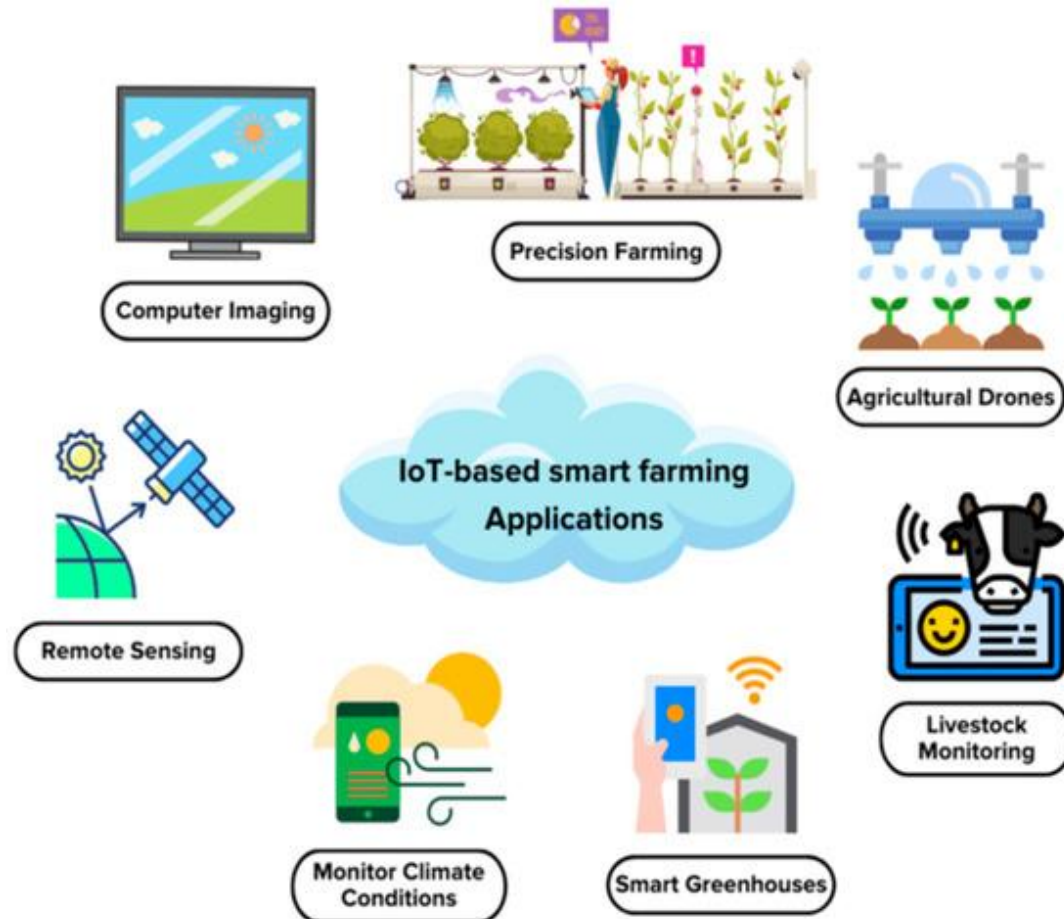




Precision Agriculture & The IoT Revolution

- **The Problem:** Traditional farming in districts like Swat and Mardan often relies on guesswork for irrigation and fertilization.
- **The IoT Solution:** Deploying sensors to measure soil moisture, pH levels, and NPK (Nitrogen, Phosphorus, Potassium) values in real-time.
- **Data-Driven Decisions:** Using this high-resolution data to provide specific "prescriptions" for each patch of an orchard, reducing waste and increasing yield.

Satellite Imagery & GIS (The View from Above)

- **Large-Scale Monitoring:** While sensors cover meters, satellites cover kilometers.
- **GeoPandas:** A specialized tool that allows us to treat geographical shapes (districts, farms) like data tables.
- **Crop Health (NDVI):** Using specific light wavelengths from satellite data to determine if a crop is healthy or stressed before the damage is visible to the human eye.

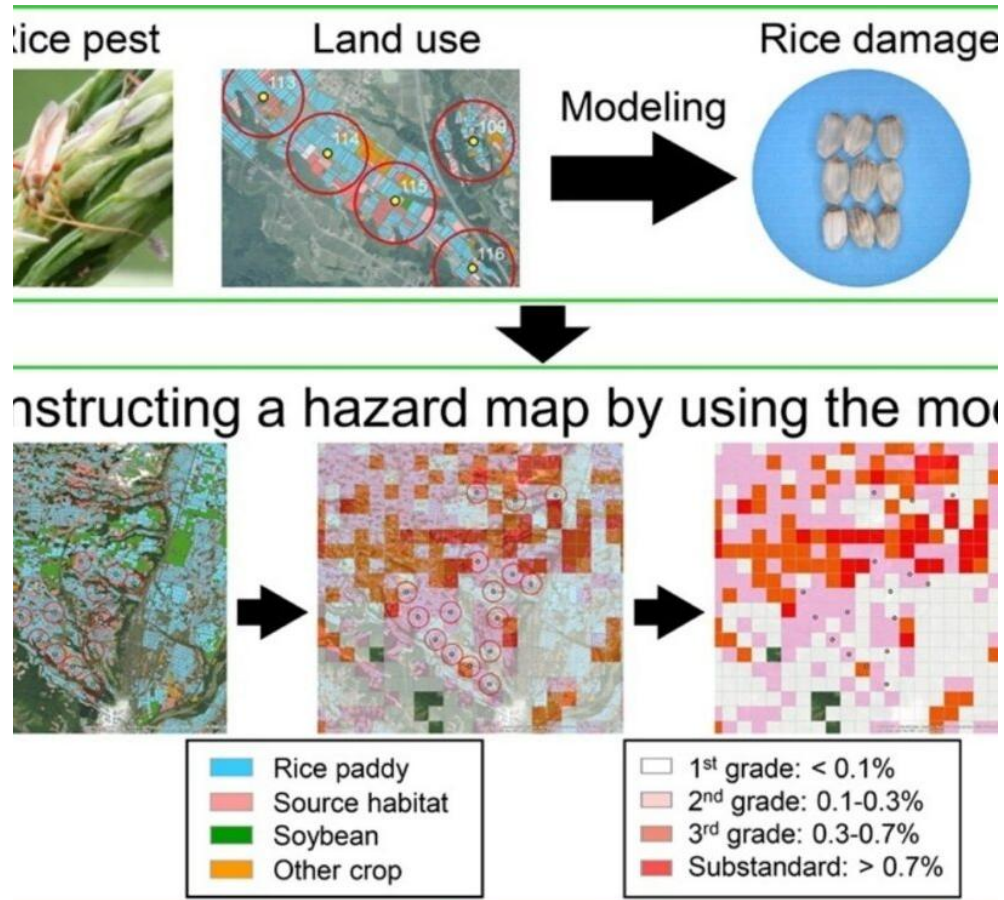


Yield Prediction & Harvest Forecasting



- **Economic Stability:** Predicting the harvest totals for major KPK crops like Wheat, Tobacco, and Sugarcane helps government planners and farmers stabilize prices.
- **The Variables:** Combining historical weather patterns (rainfall, heatwaves) with soil quality and historical harvest data.
- **Predictive Modeling:** Moving from "What happened?" to "How many tons will we harvest in June?"

Spatial Mapping of Pests and Disease



- **Tracking the Spread:** Pests like locusts or fungal diseases don't respect borders. Spatial analytics allows us to map the "front line" of an infestation.
- **Predictive Containment:** By analyzing wind patterns and humidity, we can predict which district (e.g., Charsadda vs. Nowshera) is at risk next.
- **Resource Allocation:** Sending pesticides and experts only to the areas identified as "High Risk" by the data model.